Market Power

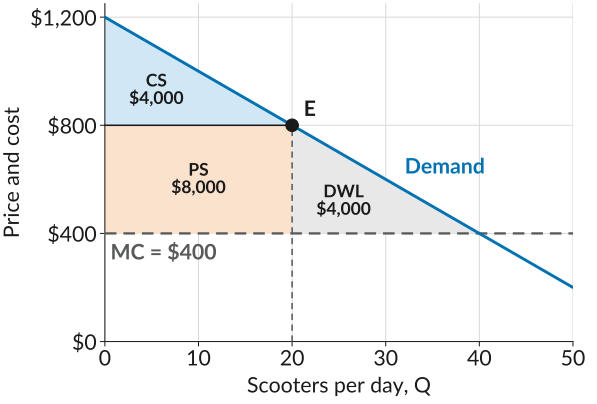
ECON 201: Principles of Microeconomics
Lecture 12

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Electric Scooters

- The company sells 20 scooters a day at \$800, at *E*. Each one costs \$400 to make.
- The three areas:

$$\text{CS} = \tfrac{1}{2} \times 20 \times 400 = 4{,}000$$
$$\text{PS} = 400 \times 20 = 8{,}000$$
$$\text{DWL} = \tfrac{1}{2} \times 20 \times 400 = 4{,}000$$



Why can this company charge more than marginal cost, and how much more?

Market Power

- The scooter company sells a product that is at least a little different from what other firms sell.
- Some buyers prefer its scooters, so they do not all switch to another brand when it charges a little more.
- If every buyer switched to the cheapest scooter, a rival could charge a little less, and this company would sell nothing at $800.
- We say a firm has **market power** if it can set its price above marginal cost without losing all its buyers to rivals.
- A firm whose product has few close **substitutes** has market power.

Price Markup

The **profit margin** is the price minus marginal cost, $P - \text{MC}$. The **price markup** is the profit margin as a share of the price:

$$\text{markup} = \frac{P - \text{MC}}{P}$$

Example: the scooter company maximizes profit by producing where MR = MC, at 20 scooters, and setting $P = 800$. Its MC is 400.

- Its profit margin is $800 - 400 = 400$ on each scooter.
- Its markup is $400/800 = 1/2$, or 50% of the price.

The Markup Rule

When a firm maximizes profit, its markup equals one over the price elasticity of demand at its chosen point:

$$\frac{P - \text{MC}}{P} = \frac{1}{\varepsilon}$$

- **Example**: at E, the scooter demand curve has slope -20, so:

$$\varepsilon = -\frac{P}{Q} \times \frac{1}{\text{slope}} = \frac{800}{20} \times \frac{1}{20} = 2$$

  - Then $1/\varepsilon = 1/2$, the same as its markup.

- The **less elastic** its demand, the **higher the markup** a firm sets.

- Fewer close substitutes make demand less elastic, so a firm with more market power sets a higher markup.

Markups in the Data

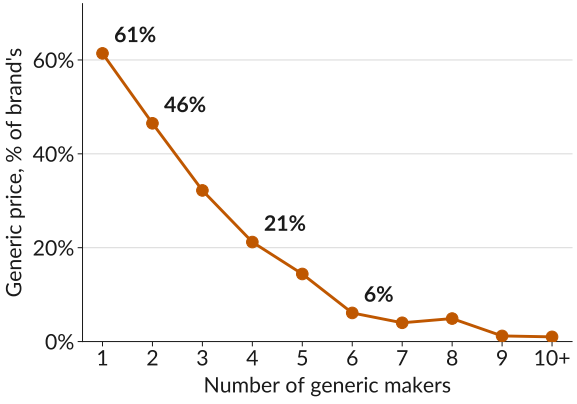
Estimates for US cars, with prices in 1983 dollars (Berry, Levinsohn, and Pakes, 1995):

Car	Price	$P - \text{MC}$	Markup	Elasticity
Mazda 323	\$5,049	\$801	16%	6.3
Nissan Sentra	\$5,661	\$880	16%	6.4
Lexus LS400	\$27,544	\$9,030	33%	3.1
BMW 735i	\$37,490	\$10,975	29%	3.4

- Small cars have many close substitutes, so their demand is elastic and their markups are low.
- Luxury cars have fewer substitutes, so their markups are about twice as high.

Generic Drugs

- When a patent runs out, other firms can sell the same drug.
- FDA data: with one generic maker, the price is 61% of the brand's. With six, it is 6%.
- More sellers of the same drug means more elastic demand and lower prices.



Where Does Market Power Come From?

A firm has market power when its buyers do not switch easily to rivals, so its demand is less elastic. We look at two sources of market power:

1. **Product differentiation**: the firm sells something different from its rivals, so buyers have fewer close substitutes.
2. **Cost advantages of size**: when producing more makes each unit cheaper, a few big firms can serve the whole market, and small rivals cannot keep up.

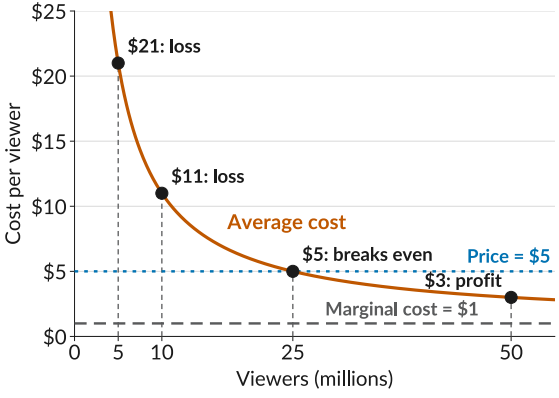
Source 1: Product Differentiation

How do firms make their products different?

1. **Location and service**: shops differ by location, selection, and extras.
2. **Innovation**: the Toyota Prius, the first mass-produced hybrid (1997), had few close rivals for about 15 years.
3. **Advertising**: it informs buyers, draws them from rivals, and builds brand loyalty.
   - US cereal makers spend 5.5% of their revenue on ads.
   - Ads raised a cereal brand's demand more than price cuts did (Shum, 2004).

Source 2: Cost Advantages of Size

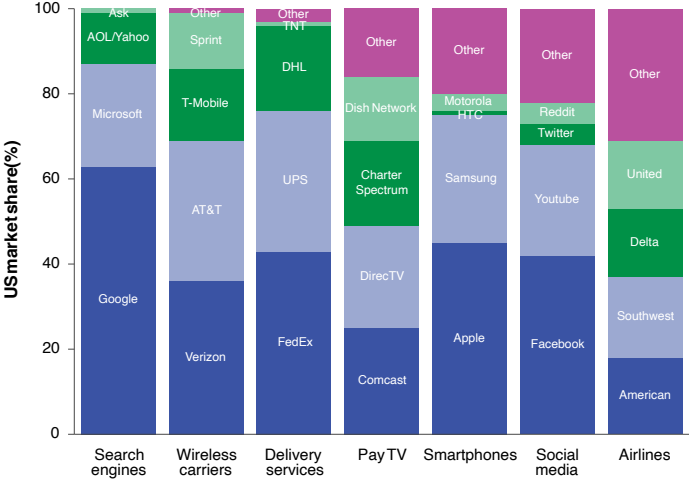
- A movie costs $100 million to make and $1 per extra stream, so AC falls as more people watch.
- A studio **breaks even**, with zero profit, where AC equals the price. At a $5 price that takes 25 million viewers; with fewer, it loses money.
- Two studios splitting 50 million viewers just break even. Three all lose. Only a few large firms survive.



Natural Monopoly

- A **monopoly** is a firm that is the only seller of a product with no close substitutes.
- When a single firm can supply the whole market at a lower AC than two firms could, the market is a **natural monopoly**: water, electricity, and gas networks, whose pipes and wires are a huge fixed cost.
- With strongly falling AC, competition can be winner-takes-all. The more searches Google ran, the more data it gathered, and the cheaper it became to serve each user.

Markets With Few Firms



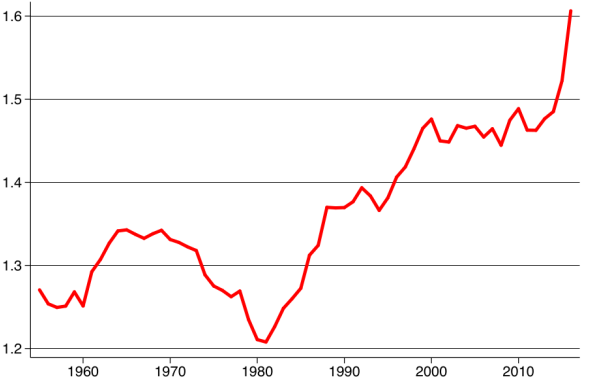
Market Structures

Economists group markets by how many firms sell, how different their products are, and how easily new firms can enter:

Structure	Firms	Products	Entry	Control over price	Example
Perfect competition	Many	Identical	Easy	None	Wheat farms
Monopolistic competition	Many	Differentiated	Easy	Some	Restaurants
Oligopoly	Few	Either	Hard	Some, rivals react	Wireless carriers
Monopoly	One	No substitutes	Blocked	Most	Water utility

Markups Over Time

- The figure shows the average P/MC of US public firms (De Loecker, Eeckhout, and Unger, 2020).
- Prices went from 21% above MC in 1980 to 61% in 2016.
- In our measure, $(P - \text{MC})/P$, that is 17% to 38%.



A price above MC means a deadweight loss. Should the government do something about rising market power?

Competition Policy

Competition policy (or **antitrust policy**) is government policy and law that limits market power, prevents cartels, and regulates how firms compete.

- Authorities investigate when one or a few dominant firms are limiting their rivals, so buyers get less from competition than they could.
- They step in only when a firm has both the ability and the incentive to abuse its market power.
- Firms limit competition in two ways: by keeping rivals out, and by making their own demand less elastic.

Keeping Rivals Out

- From expanding: UK supermarkets bought land near their stores and left it empty, so rivals could not build there.
- From inputs: Ferrero, the maker of Nutella, bought one of its hazelnut suppliers. Regulators allowed it, since enough other suppliers remained.
- From ideas: a **patent** gives a firm the sole right to an idea for a time, as a reward for research. When it expires, rivals enter: generic drugs.
- From buyers: Microsoft kept rival web browsers off Windows, so they could not reach its users. After years of investigation and fines, a browser market developed.

Mergers, Cartels, and Loyalty

- Merging with rivals: Meta bought Instagram and WhatsApp. Authorities weigh the harm against the savings from scale and network effects, and have sometimes let such mergers through.
- Colluding: a **cartel** is a group of firms that agree to act like one monopoly. Cartels are illegal in most countries, but hard to prove.
- Making buyers less price-sensitive: advertising that builds a brand, paying for shelf space or the top spot in a search, and products that work best together, like Apple's.

Case by Case

- Authorities weigh the harms and the benefits of each case on its own.
- For a natural monopoly, the answer is usually close regulation of its prices, or public ownership.
- Platforms are a dilemma: scale and network effects make them look like natural monopolies, yet many sellers and advertisers compete on them. The debate continues.
- Competition keeps moving, and policy often plays catch-up: Chrome now dominates the browser market that the Microsoft case opened up.

What to Do Next

- Before next class: read sections **7.8**, **7.9**, **7.11**, and **7.12**, the markup rule in **7.6**, and the **notes on market structures**. Go over the **slides**.
- We skip CORE's derivation of the markup rule from isoprofit curves, and its discussion of digital platforms.
- Work through the **practice problems** for this lecture.
- Next class: with only a few firms, each firm's best price depends on what its rivals charge. We model this as a game.